Solve the 3-story 3×2-span 3D spatial frame structure problem using the finite element method.

Problem Details:
Geometry:
- Type: 3-story 3×2-span spatial rigid frame (X: left-right, Y: front-back, Z: vertical).
- Span/height: X-direction: 3 spans × 6000mm; Y-direction: 2 spans × 3000mm; Z-direction: 3 floors × 3000mm.
- Sections: Columns: 500mm square section. X-beams: b=Y-axis width=500mm, h=Z-axis height=500mm. Y-beams: b=X-axis width=500mm, h=Z-axis height=500mm.
- Material: Linear elastic, E=2.06×10e5MPa, ν=0.2. No body forces and consider shear deformation if necessary.

Nodal Coordinates (X, Y, Z, unit: mm):
Total 48 nodes (4 positions in X, 3 in Y, 4 in Z), with coordinates determined by spans/heights:
- X-direction: 3 spans × 6000 → X = 0, 6000, 12000, 18000 (4 positions: i=1, 2, 3, 4)
- Y-direction: 2 spans × 3000 → Y = 0, 3000, 6000 (3 positions: j=1, 2, 3)
- Z-direction: 3 floors × 3000 → Z = 0, 3000, 6000, 9000 (4 positions: k=1, 2, 3, 4)

Element Division:
Total elements: 87 (36 columns + 27 X-beams + 24 Y-beams).
Z-direction Columns (vertical, Z-direction): Connect nodes (i, j, k) → nodes (i, j, k+1).
- Quantity: 4 (X) × 3 (Y) × 3 (Z-intervals: k=1→2, 2→3, 3→4) = 36.

X-direction beams (horizontal, X-direction): Connect nodes (i, j, k) → nodes (i+1, j, k).
- Quantity: 3 (X-intervals: i=1→2, 2→3, 3→4) × 3 (Y) × 3 (floors: k=2, 3, 4) = 27.

Y-direction beams (horizontal, Y-direction): Connect nodes (i, j, k) → nodes (i, j+1, k).
- Quantity: 4 (X) × 2 (Y-intervals: j=1→2, 2→3) × 3 (floors: k=2, 3, 4) = 24.

Boundary Conditions:
Boundaries: Bottom nodes: fully fixed (ux=uy=uz=0; θx=θy=θz=0); Other nodes with rigid connections (consistent displacements/rotations).
Loads:
- Uniform Z-direction load (p=-20N/mm) along the length of all beams (both X and Y directions);
- Uniform X-direction load (w=300N/mm) applied from bottom to top of the columns where X=0, Z∈ [0,9000].

Calculations required:
- Nodal Displacement and rotation: ux, uy, uz, θx, θy, and θz;
- For each element (beams and columns), output the displacements and rotations (ux, uy, uz, θx, θy, θz) at points along the element axis every 200mm;
- Provide visualization of the deformation diagram.

Programming language: Python 3.10.
Permitted libraries: NumPy 2.1.1, SciPy 1.14.1, JAX 0.4.30, Matplotlib 3.9.2.